

AI Endpoint Intelligence Platform - Detailed Project Plan

Vision: Build an enterprise AI platform that continuously monitors endpoints, predicts failures, explains root causes, automates remediation, and provides fleet-wide intelligence.

1. Objectives

- Predict hardware/software failures before downtime.
- Provide AI-powered IT Copilot.
- Automate troubleshooting and approved repairs.
- Deliver enterprise fleet monitoring.
- Support Windows first, then Linux/macOS.

2. Core Modules

1. Desktop Telemetry Agent
2. Telemetry Collector API
3. Time-Series Database
4. Machine Learning Engine
5. Rule Engine
6. AI Orchestrator
7. IT Copilot
8. Root Cause Agent
9. Auto Repair Agent
10. Fleet Intelligence Dashboard

3. Technology Stack

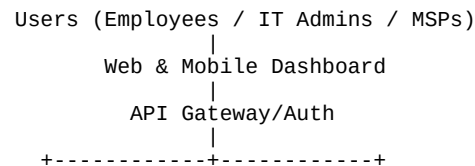
Desktop Agent: Rust/C#
Backend: FastAPI
Database: PostgreSQL + TimescaleDB
Queue: Kafka/RabbitMQ
ML: PyTorch, XGBoost
LLM: GPT-5.5 / Llama
Vector DB: Qdrant
Frontend: React + Next.js
Deployment: Docker + Kubernetes

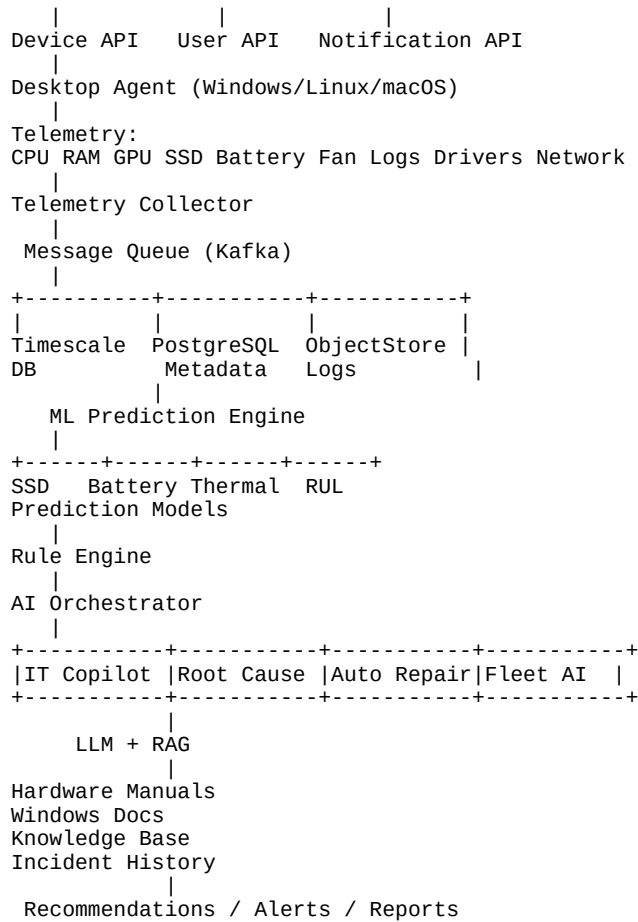
4. Development Phases

Phase 1: Agent, API, Dashboard, Health Score
Phase 2: Prediction Models (SSD, Battery, Thermal)
Phase 3: AI Copilot + RAG
Phase 4: Fleet Intelligence + Auto Repair
Phase 5: Enterprise Scaling & Continuous Learning

5. Enterprise Architecture

DETAILED ARCHITECTURE





6. AI Workflow

1. Collect telemetry from endpoints.
2. Store historical metrics.
3. Detect anomalies.
4. Predict failures using ML.
5. Correlate logs and telemetry.
6. Generate root cause.
7. AI Copilot explains issue.
8. Auto Repair executes approved fixes.
9. Fleet dashboard aggregates organization health.
10. Feedback loop retrains models.

7. Expected Outcomes

- Reduce downtime
- Lower maintenance costs
- Extend hardware lifespan
- Improve IT productivity
- Create recurring SaaS revenue